

MIAE 221 · Rapid Review Sheet · Part 6

Mechanical Properties of Metals

1. Stress, Strain & Elasticity

- **Engineering Stress:** $\sigma = \frac{F}{A_0}$ (MPa = N/mm² or GPa = 10³ MPa).
 - **Engineering Strain:** $\epsilon = \frac{\Delta l}{l_0} = \frac{l_i - l_0}{l_0}$ (dimensionless or %).
 - **Hooke's Law (Elasticity):** $\sigma = E \cdot \epsilon \implies E = \frac{\Delta \sigma}{\Delta \epsilon}$ (Young's Modulus, measure of stiffness).
 - **Atomic Origin of E :** $E \propto \left(\frac{dF}{dr} \right)_{r_0}$
 $\left(\frac{d^2 U}{dr^2} \right)_{r_0}$ (curvature of interatomic potential well). Deep, steep well implies High E , High T_m , Low α .
 - **Poisson's Ratio:** $\nu = -\frac{\epsilon_{\text{lateral}}}{\epsilon_{\text{axial}}} = -\frac{\Delta d/d_0}{\Delta l/l_0}$ (typically 0.25 – 0.35 for metals).
 - **Shear Modulus:** $G = \frac{E}{2(1 + \nu)}$ ($\tau = G \cdot \gamma$).
-

2. Plastic Deformation, Yielding & Tensile Strength

- **Yield Strength (σ_y):** Standardized by the 0.002 (0.2) **strain offset** line parallel to E : $\sigma = E(\epsilon - 0.002)$.
 - **Upper & Lower Yield Point:** In annealed low-carbon steel due to interstitial carbon/nitrogen pinning (Cottrell atmospheres $\rightarrow$ Lüders bands). Lower yield is used for design.
 - **Ultimate Tensile Strength (σ_{UTS}):** Maximum engineering stress on curve: $\sigma_{UTS} = \frac{F_{\max}}{A_0}$. Marks the onset of **necking**.
-

3. Ductility, Resilience & Toughness

- **Percent Elongation:**
- **Percent Reduction in Area:** (square diameters!).
- **Modulus of Resilience (U_r):** Elastic energy absorbed up to yielding:

$$U_r = \frac{1}{2} \sigma_y \epsilon_y = \frac{\sigma_y^2}{2E} \quad (\text{J/m}^3 \text{ or kJ/m}^3)$$

- **Toughness:** Total area under the complete $\sigma - \epsilon$ curve up to fracture. Requires combination of strength AND ductility.
-

4. True Stress-Strain & Strain Hardening

- **True Stress:** $\sigma_T = \frac{F}{A_i} = \sigma(1 + \epsilon)$ (valid prior to necking).
 - **True Strain:** $\epsilon_T = \ln\left(\frac{l_i}{l_0}\right) = \ln(1 + \epsilon)$ (valid prior to necking).
 - **Hollomon Power Law:** $\sigma_T = K \cdot \epsilon_T^n$ (n = strain hardening exponent, 0.10 – 0.50).
 - **Necking Criterion:** At UTS, $\frac{d\sigma_T}{d\epsilon_T} = \sigma_T \implies \epsilon_T = n$.
-

5. Hardness Testing Reference

- **Brinell (HB):** Steel/tungsten ball ($D = 10$ mm):

$$HB = \frac{2P}{\pi D (D - \sqrt{D^2 - d^2})}$$

- **Rockwell:** Diamond cone (120°) or steel ball ($1/16''$). Read depth directly: **HRA** (cemented carbides), **HRB** (brass/Al), **HRC** (hardened steels).
- **Vickers (HV):** Diamond pyramid (136°). Micro/macro hardness on a single scale.
- **Knoop (HK):** Microhardness for thin coatings and brittle ceramics.
- **Empirical Correlation:** $TS, (\text{MPa}) \approx 3.45 \times HB$.